

BatteryMonitor Windows

User Manual — v1.0

Overview

BatteryMonitor displays real-time battery data in two side-by-side charts and a two-row header. Controls are in the footer bar at the bottom.

Installation & Requirements

Requirements

- **Operating system:** Windows 10 or Windows 11
- **.NET Framework 4.0** or later (included in Windows 10/11 by default)
- **Administrator privileges** — required for WMI battery queries (health, cycle count)
- **TC66 USB meter** (optional) — for external V/i measurement and charging efficiency

Installation

- Download the latest release from the Releases page on GitHub
- Extract the ZIP — no installer required
- Ensure **app.manifest** is in the same folder as **BatteryMonitor.exe**
- Run **BatteryMonitor.exe** — accept the UAC prompt to grant administrator access
- If Windows Smart App Control blocks the executable, right-click → Properties → Unblock

First Run

- Click **Start** to begin recording
- To use a TC66 meter, select the correct COM port and click **Con**

The TC66 USB Power Meter (Optional)

BatteryMonitor optionally integrates with the **MakerHawk TC66 USB Power Meter** (~\$26.99 on Amazon: *MakerHawk USB Power Meter, TC66 USB Tester Type C, USB Voltage Meter and Current Tester, 0.96" IPS Color LCD Display, PD Ammeter Voltmeter QC 2.0/3.0*). The TC66 measures voltage, current, and power independently of the BMS, enabling accurate DC-DC charging efficiency calculations. It connects via USB serial (typically COM3–COM6 on Windows) and streams data to BatteryMonitor while displaying live readings on its own color LCD.

■ **Note:** To reset the TC66's cumulative capacity (Q_{in}) and energy (E_{in}) counters to zero, the physical button on the meter must be pressed. There is no software reset available.

Footer Controls (left to right)

- **Start / Stop** — Begins or ends a recording session. The system is kept awake while recording.
- **Clear** — Clears all chart data and resets all accumulators without stopping the session.

- **Load CSV** — Loads a previously saved CSV file for review. Live recording is suspended.
- **Int: N s** — Sample interval in seconds (1–60). Lower values give finer resolution but larger files.
- **Stop** (red, TC66 section) — Disconnects the TC66 meter.
- **Span:** — Time window shown in both charts (All, 1h, 2h, 4h, 8h). Does not affect recording.
- **Refr** — Forces an immediate TC66 screen refresh.
- **TC66: COMx** — Serial port selector for the TC66 USB meter.
- **Disc / Con** — Connects or disconnects the TC66 meter on the selected port.
- **■** — Flips the TC66 display orientation (useful when the meter is inverted).
- **Psys: N W** — System baseline power subtracted from TC66 input when computing CE. Typically 4–5 W.
- **LR pts: N** — Minimum SOC points for the auto linear regression (TT5 prediction).
- **Run time** — Elapsed time since Start was pressed.
- **N samples** — Total samples recorded in the current session.

Header — Row 0 (BMS Data)

Row 0 is always visible. The **status label** on the left shows the current state:

Label	Meaning
CHRG	Charging
DCHRG	Discharging
IDLE	No significant current flow
FULL	Fully charged (trickle only)

Fields (discharge and charge)

Field	Description	Format
U=	Battery terminal voltage (V)	F2 always
i=	Current (A) — negative when discharging	F2 always
R=	Internal resistance (Ω)	Integer
P=	Power (W) — negative when discharging	F2 if <10, F1 if ≥ 10
Qc=	Segment charge since last state transition (Ah)	F2
Ec=	Segment energy since last state transition (Wh)	F2 if <10, F1 if ≥ 10
Qt=	Cumulative charge since Start (Ah)	F2
Et=	Cumulative energy since Start (Wh)	F2 if <10, F1 if ≥ 10
SOC=	State of charge reported by BMS (%)	F1
SoH=	State of health: full-charge $\div$ design capacity (%)	F1

Header — Row 1 (TC66 Data + Computed)

Row 1 is visible when the TC66 is connected or when the TT5 countdown is active.

When TC66 is connected

Field	Description
U=	TC66 measured voltage (V)
i=	TC66 measured current (A)
R=	TC66 computed resistance (Ω), integer
P=	TC66 measured power (W)
Qin=	Cumulative charge delivered by TC66 since connection (Ah)
Ein=	Cumulative energy delivered by TC66 since connection (Wh)
CE=	Charging efficiency: $E_c \div (E_{in} - P_{sys} \times t) \times 100$ (%)
T.TC=	TC66 internal temperature ($^{\circ}\text{C}$) — not battery cell temperature

Always visible during discharge

Field	Description
TT5:	Estimated time to 5% SOC from auto linear regression on recent SOC data
Cyc=	Battery cycle count reported by BMS

The Two Charts

BatteryMonitor displays two dual-axis charts side by side, updated in real time. Both share the same elapsed-time X axis. Curve colors switch automatically between charge (green tones), discharge (red/orange tones), and idle (gray).

Chart 1 — Voltage, Current & Capacity

Left Y axis (U, V): battery and TC66 voltage.

Right Y axis (i, A & Qc, Ah): current and cumulative segment capacity.

Curve	Style	Description
i, A	Solid	BMS discharge/charge current (negative = discharge)
i.TC, A	Dashed	TC66 measured current
U, V	Solid	BMS battery terminal voltage
U.TC, V	Dashed	TC66 measured input voltage
Qc, Ah	Solid	Segment charge (resets on state transition)
Qt, Ah	Solid	Cumulative charge since Start

TC66 curves shown only when meter is connected.

Chart 2 — SOC, Power & Temperature

Left Y axis (P, W & Energy, Wh): power and cumulative energy.

Right Y axis (SOC, % & T, °C): state of charge and TC66 temperature.

Curve	Style	Description
SOC, %	Solid	State of charge reported by BMS
P, W	Solid	BMS power (negative = discharge)
P.TC, W	Dashed	TC66 input power
Ec, Wh	Dashed	Segment energy since last state transition
Et, Wh	Solid	Cumulative energy since Start
T.TC, °C	Dashed	TC66 internal temperature

Interactive Chart Features

Running Average (2-click, both charts)

Click two points on any curve to display a running average line and value badge.

- Click 1: sets the start point
- Click 2: sets the end point; the average value is displayed
- **Click to clear avg** (top-left of chart): removes the average

Linear Regression — TT5 Prediction (3-click, Chart 2 only)

The **auto** regression continuously fits recent SOC data and projects to 5% SOC for the TT5 estimate. A **manual** 3-click regression is also available:

- Click 1 & 2: define the SOC data range to fit
- Click 3: sets the target SOC; regression line and intercept are displayed
- **Click to clear LR** (top-right of chart): removes the regression line

Regression line color:

- **Green:** $R^2 \geq 0.90$ — reliable prediction
- **Orange:** $R^2 < 0.90$ — poor fit; treat estimate with caution

Cursor Tooltip

Hovering over either chart shows the time and value of the nearest data point.

Legend

Click the legend to cycle through three positions (top-left, center, bottom-left).

Time Window (Span)

Controls the time window in both charts (All, 1h, 2h, 4h, 8h). Does not affect recording.

Alerts

- **Charge complete:** when charging current drops to zero (CHRG→IDLE), a triple beep (880 Hz) sounds every 60 seconds for up to 10 minutes. Dismissed by the red **X Beep** button on the footer right.
- **Low SOC:** a single beep (660 Hz) sounds at 10% SOC during discharge, then at each subsequent 1% step down to 1%.

CSV File Format

Files named **BatteryLog_YYYYMMDD_HHMMSS.csv**, saved to the application folder, auto-saved every 10 seconds. TC66 columns always present; blank when meter not connected.

Column	Unit	Format	Description
Timestamp	—	yyyy-MM -dd HH: mm:ss	Wall-clock time of sample
Elapsed_s	s	F1	Elapsed time since Start
Voltage_V	V	F3	BMS battery terminal voltage
Current_A	A	F3	BMS current (negative = discharge)
Power_W	W	F2	BMS power (negative = discharge)
SOC_Pct	%	F1	State of charge
State	—	text	Charging / Discharging / Idle / Full
Qt_Ah	Ah	F6	Cumulative charge since Start

Et_Wh	Wh	F3	Cumulative energy since Start
Qc_Ah	Ah	F6	Segment charge since last state transition
Ec_Wh	Wh	F3	Segment energy since last state transition
TC66_V	V	F4	TC66 voltage (blank if no TC66)
TC66_A	A	F5	TC66 current (blank if no TC66)
TC66_W	W	F4	TC66 power (blank if no TC66)
TC66_Temp	°C	integer	TC66 internal temperature (blank if no TC66)
TC66_Ah	Ah	F6	TC66 cumulative charge (blank if no TC66)
TC66_Wh	Wh	F3	TC66 cumulative energy (blank if no TC66)

Known BMS Limitations

- **SOC step size varies by model:** reported in discrete steps; size depends on laptop model and SOC level, often smaller near full discharge.
- **BMS holdback (model-specific):** on the ThinkPad T490, SOC was observed to hold at 7% for 1–2 minutes before dropping sharply — a deliberate BMS algorithm, not a measurement artifact. Affects TT5 accuracy in this range.
- **Battery cell temperature** not accessible via standard Windows API on tested Lenovo models (Yoga 7 2-in-1, ThinkPad T490, ThinkPad X1 Carbon). **T.TC=** shows the TC66 meter's own sensor temperature.

About

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Repository: <https://github.com/ASG49/BatteryMonitor-Windows>

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